Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/17/2024
PI Name:	Iyer, Janani Ph.D.		
Project Title:	Sex-Specific Physiological and Transcriptomic CNS Responses to Combined Effects of Spaceflight Stressors in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	05/01/2023	**End Date:**	04/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2025 per F. Hernandez/ARC. Previous end date was 3/15/24 (Ed., 3/15/24).		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Ronca, April Ph.D. (NASA Ames Research Center) Mhatre, Siddhita Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During space exploration, damage to the central nervous system (CNS) due to altered gravity is a significant risk, along with the constant exposure to elevated CO2 levels. The combination of these stressors can negatively impact the CNS health that may lead to decrements in astronaut performance, posing a risk to the crew and the mission. Thus, there is an unmet need to unravel the mechanisms and pathways affected by these combined spaceflight stressors. In this proposal, we aim to address sex-specific and long-term responses to spaceflight stressors (mimicking the longitudinal post-flight evaluations in astronauts). We will perform behavioral, brain morphological, and biochemical assays, along with cell-specific transcriptomic profiling in Drosophila melanogaster to investigate the underlying mechanistic responses to single and combined exposures of altered gravity and elevated CO2. While this solicitation is limited in time and scope, we anticipate that our findings from this study will inform future investigations in vertebrate models and contribute to new research that will address the biomedical outcomes of deep space stressors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effects of gravity as a continuum on the central nervous system (CNS) are less explored. Also, even though elevated CO2 has been a long-known stressor in spaceflight, data on its effects on different systems, specifically CNS, is very scant. Further, the combined effects of altered gravity and elevated CO2 are unknown. This study will focus on the neuronal responses to oxidative stress induced by altered gravity and elevated CO2. Additionally, the proposal offers a unique opportunity to evaluate key behavioral, biochemical, cellular, morphological, and molecular responses longitudinally over multiple time points post-exposure to the stressors, which are currently not well-characterized. The genetic basis of responses to oxidative stress in Drosophila and vertebrates is conserved, thus this study will serve as the first step in identifying pathways and mechanisms that can be used for therapeutic intervention. Neurological Disorders: Studies suggest that elevated CO2 levels may exacerbate symptoms in individuals with neurological disorders such as Alzheimer's disease and Parkinson's disease. Research in this area helps elucidate the mechanisms through which CO2 affects neurological function and may provide insights into potential therapeutic interventions to mitigate these effects and improve the quality of life for affected individuals.
Task Progress:	Ground-based studies, such as hypergravity (HG) using centrifugation, offer a cost-effective and reliable means to simulate gravitational conditions found on Earth. Previous research utilizing animal spaceflight and HG studies has provided valuable insights into the impact of altered gravity on behavioral and neural functions. In this study, we aim to explore dose-dependent responses to altered gravity by comparing central nervous system (CNS) reactions under two different gravitational loads: 1.2g and 3g. Additionally, we seek to investigate the effects of elevated CO2 levels, which are commonly observed in closed space habitats like the International Space Station (ISS) due to technical challenges in CO2 scrubbing. Recent literature and our preliminary investigations indicate that prolonged exposure to elevated CO2 can influence CNS physiology, immune response, and key metabolic pathways such as oxidative phosphorylation. Astronauts experience constant exposure to altered gravity (AG) and elevated CO2 levels during spaceflight, yet the combined effects of AG and elevated CO2 on CNS remain poorly understood. Our study is designed to address this gap by subjecting adult flies aged 3 days to chronic HG (1.2g and 3g) and/or elevated CO2 (4000 ppm) for 15 days. The selection of 1.2g and 3g as gravitational loads were based on preliminary findings, with 3g chosen to evaluate CNS responses across a gravity continuum. The CO2 concentration of 4000 ppm reflects average levels observed on the ISS during our MVP-Fly-01 spaceflight mission. Our treatment paradigm allows for the longitudinal study of CNS responses to altered gravity, with data collection at immediate post-treatment (R+0), five days post-treatment (R+5), ten days post-treatment (R+10), and twenty-five days post-treatment (R+25). This comprehensive approach will provide valuable insights into the combined effects of altered gravity and elevated CO2 on CNS function, shedding light on the challenges faced by astronauts during space missions and informing strategies to mitigate potential adverse effects. Neurobehavioral changes were examined under both single and combinatorial stressor conditions using the climbing assay, which manipulates the negative geotaxis reflex in Drosophila melanogaster. At the R+0 timepoint, dose-dependent alterations in climbing ability were evident in both male and female flies, indicating dysfunction in locomotive pathways within the brain-body axis. By the R+5 timepoint, full acclimation to both 1.2g and 3g treatments was observed, with climbing ability restored in both sexes. However, in the case of elevated CO2 exposure, males exhibited a trend of reduced climbing ability at R+0, though not statistically significant between 1g and 1gCO2 males. Notably, 3gCO2 males displayed a statistically significant decrease in climbing ability ($p=0.0033$). Conversely, females showed a dose-dependent response, with 3gCO2 females exhibiting significantly greater climbing deficits compared to 1gCO2 and 1g controls ($p<0.0001$). Ongoing investigation will evaluate behavioral responses at subsequent longitudinal time points (R+5, R+10, and R+25) under both single and combinatorial stressor conditions, providing further insights into the persistence and evolution of neurobehavioral changes in response to altered gravity and elevated CO2 levels. In addition to behavioral assessments, bioenergetic assays were conducted to evaluate various parameters including mitochondrial membrane potential (MMP), adenosine triphosphate (ATP) quantification, Reactive Oxygen Species (ROS) levels, and mitotracker (indicative of mitochondrial abundance). Preliminary findings from these assays indicate notable alterations in ROS levels, mitochondrial membrane potential, and ATP production. Specifically, increased ROS levels suggest heightened oxidative stress within the cellular environment. Additionally, changes in mitochondrial membrane potential indicate potential disruptions in mitochondrial function, which can impact cellular energy metabolism and overall cellular health. Moreover, variations in ATP production suggest alterations in cellular energy production pathways, which may reflect adaptive responses to the imposed stressors. Further analysis and validation of these findings will provide deeper insights into the cellular responses to altered gravity and elevated CO2 levels. Understanding the bioenergetic implications of these stressors is critical for elucidating their effects on cellular physiology and may inform strategies for mitigating potential adverse outcomes in organisms exposed to such environmental conditions. Our preliminary findings indicate an increase in dopamine (DA) neuron loss and elevated cell death in chronic hypergravity (HG) conditions and spaceflight. Additionally, both spaceflight-exposed and terrestrial flies exposed to elevated CO2 demonstrate heightened oxidative stress, as evidenced by increased 8-oxo-dG puncta in the brain. To further investigate the effects of HG with or without CO2 on brain structure, we employ multiple GAL4 fly lines for brain morphology assessment. In all fly lines, we assess cell death and oxidative damage using anti-CC3 antibody for cell death detection and anti-8-oxo-dG antibody for oxidative damage assessment. At the R+0 timepoint, brain dissections and staining have been completed for all fly lines. The samples are currently at various stages of imaging (confocal) and subsequent analysis. Additionally, brain dissections for other time points have been conducted and are

	ready for further analysis. This comprehensive approach allows us to investigate the structural and cellular changes in the brain in response to HG and elevated CO2 exposure, providing insights into the underlying mechanisms of neurobiological adaptation and potential impacts on cognitive function and neuronal health.
Bibliography Type:	Description: (Last Updated: 04/08/2025)
Abstracts for Journals and Proceedings	Prabhu A. "Sex-specific CNS responses to combined effects of spaceflight stressors in Drosophila melanogaster." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Wyneimko T. "Central nervous system responses to gravity as a continuum in Drosophila melanogaster." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023